

**What makes an
asset the
dominant
intermediary?**



**Centre for Blockchain
Technologies**

The Making of Dominant Vehicle Currencies

Evidence from DeFi
Jiahua Xu

How we reconstruct a route from one transaction

Direct swap

one transaction



Sequential vehicle route

one transaction



Parallel split

one transaction



Round trip

one transaction



Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k .
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.



Count-weighted dominance

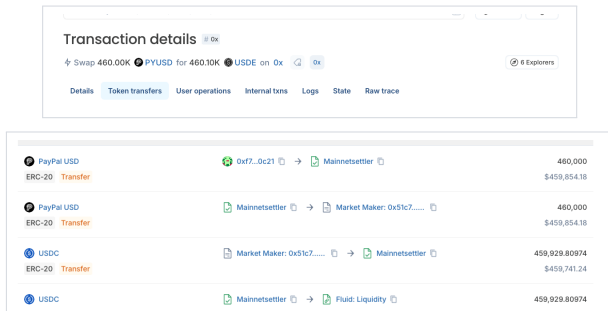
Share of indirect routes that pass through k

Value-weighted dominance

Share of comparable routed value that passes through k

The transaction starts with PYUSD

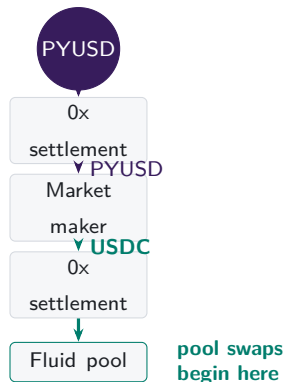
EXPLORER RECORD



The screenshot shows a transaction details page for a swap of 460.00K PYUSD for 460.10K USDE on 0x. Below this, a table lists four token transfers:

Token	From	To	Amount	Value
PayPal USD (ERC-20)	0xf7...0c21	Mainnetsettlr	460,000	\$459,854.18
PayPal USD (ERC-20)	Mainnetsettlr	Market Maker: 0x51c7.....	460,000	\$459,854.18
USDC (ERC-20)	Market Maker: 0x51c7.....	Mainnetsettlr	459,929.80974	\$459,741.24
USDC	Mainnetsettlr	Fluid: Liquidity	459,929.80974	

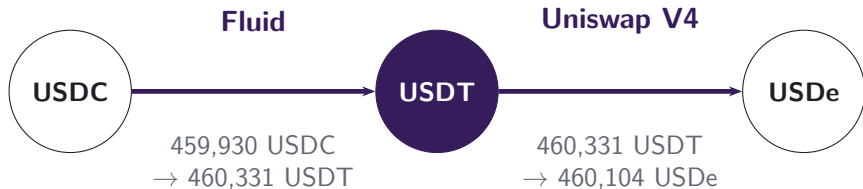
SETTLEMENT HANDOFF



PYUSD reaches 0x settlement; the market maker supplies USDC for execution in the pools.

Note: The explorer screenshot and transfer trace refer to transaction 0xbda1d07f...bf67ad9 on 10 January 2026.

USDT is the intermediary in the connected pool route



vehicle

The connected component converts USDC into USDe across two pools.
USDT is the vehicle because it links the two legs.

Note: The route is the connected pool-swap component of transaction 0xbda1d07f...bf67ad9 on 10 January 2026.

Architecture changes routing opportunities

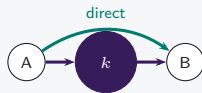
V1 mandatory hub



Only ETH-token pools: token-to-token execution must pass through ETH.

Hub use is imposed

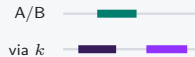
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

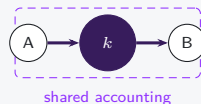
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



Two pools still execute; intermediate token movement may be netted.

Settlement and vehicle use can diverge

Architecture feasible rules

Exposure realised pool and route use

Calendar date timing only

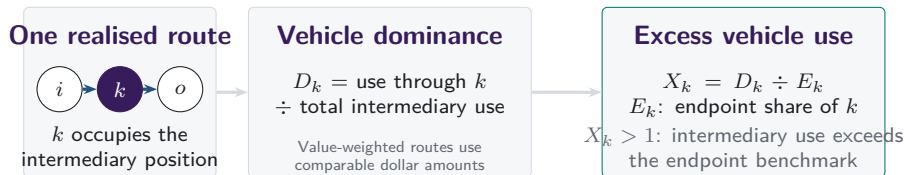
The sample spans Uniswap V1 through V4

DATA

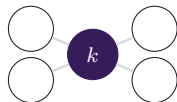


Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

Vehicle dominance aggregates realised intermediary choices

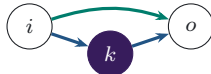


Separate characteristics of the asset and route set



Network position

position of k in the feasible graph

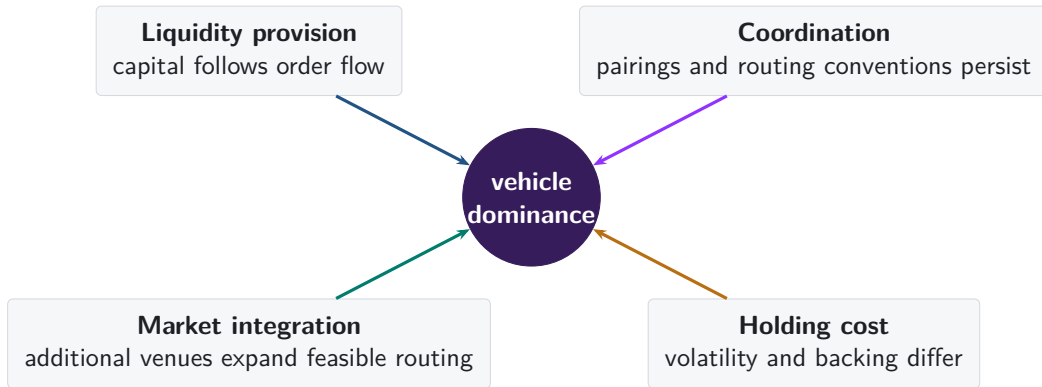


Execution cost

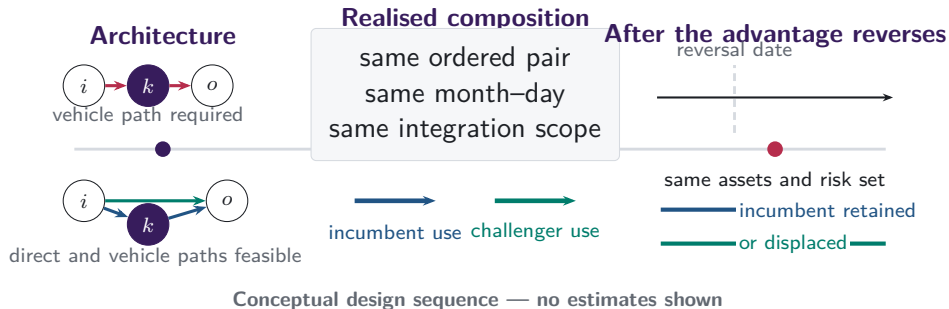
which route returns more at the same state

Endpoint demand supplies the benchmark; network position and same-state route costs remain separate characteristics.

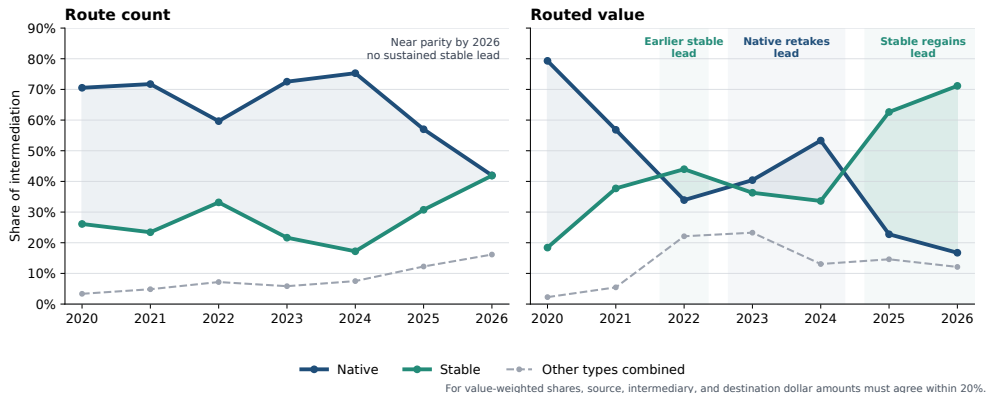
Why might one asset become the dominant vehicle?



Architecture opens routes; realised trades reveal which vehicle is used

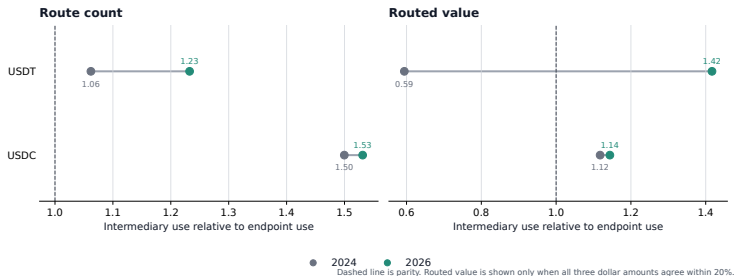


Stablecoins regain the lead in routed value



Note: Annual shares of observed vehicle routes. The 2026 observation covers January–June; Other combines imported, liquid-staking, and residual assets.

USDT drives the rise in stablecoin intermediation

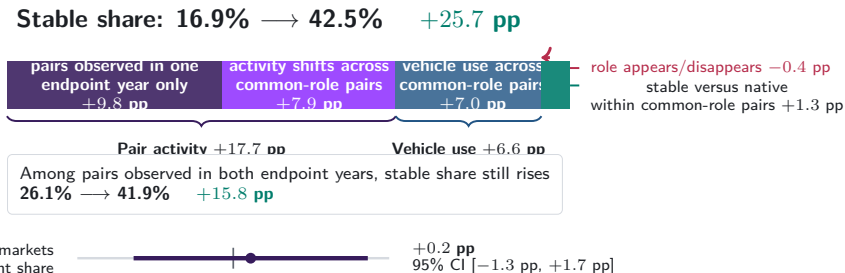


USDT crosses parity by routed value and extends its count advantage; USDC changes little.

93.7% of the count-share change and 85.6% of the routed-value change occur within the single- and cross-venue categories.

Note: Excess use is intermediary share divided by endpoint share on the same route perimeter. The comparison uses the same January–June dates in 2024 and 2026.

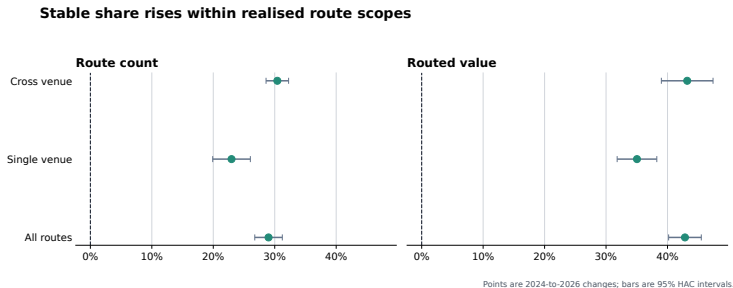
Pair activity and vehicle use: 94.9% of the gain



Dollar-weighted routes show the same broad allocation. Stable share rises +42.8 pp, with +26.2 pp from activity weights among continuing pairs and only 0.0 pp from stable-for-native substitution within them.

Note: The sample contains exact two-leg routes observed in 2024 and January–June 2026. Matched estimates compare the same ordered endpoint pair, calendar date, and single- versus cross-venue route category. Dollar-weighted calculations retain routes for which the source, intermediary, and destination dollar amounts agree within 20%.

Stablecoin use rises within each venue scope

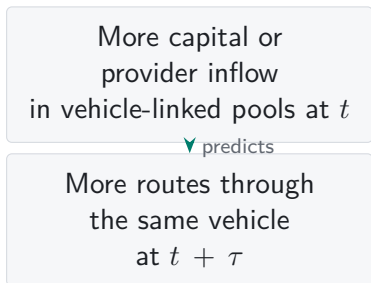


The increase appears within both realised route scopes, so a change in the single- versus cross-venue mix cannot account for the aggregate rotation. Cross-venue status remains selected; the comparison does not identify an effect of integration, routing software, or protocol architecture.

Note: Changes compare the same January–June dates in 2024 and 2026. Bars show 95% calendar-HAC confidence intervals.

Does liquidity move before vehicle use or follow it?

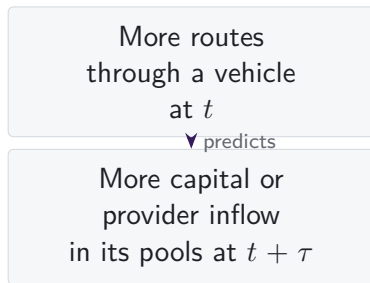
LIQUIDITY PRECEDES USE



Liquidity supply may help a vehicle attract later routes.

V2 reserve-based capital and V3 mint/burn flows are measured separately at exact calendar lags. Either direction may also reflect common demand, volatility, or routing shocks.

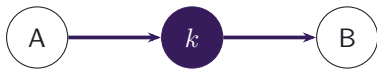
USE PRECEDES LIQUIDITY



Providers may instead follow routed demand.

V4 separates the economic route from physical settlement

ECONOMIC ROUTE



Pool calls route
through intermediary k .

V4 may net the token movements
used to settle those calls.

COMPARISON

- Compare routes in the same week, endpoint pair, intermediary, direction, and trade-size group.
- Record route use and token transfers separately.
- Compare feasible alternatives before interpreting vehicle choice.

The architecture can make a physical transfer unnecessary. The empirical test asks whether transfers fall among otherwise comparable routes.

Challenger cost advantage predicts subsequent vehicle share

30-day outcome	$\hat{\beta}$	SE	t
Change in challenger vehicle share	4.825	0.366	13.17
Change in incumbent vehicle share	-3.872	0.420	-9.22
Challenger leads after 30 days	9.947	0.486	20.48

Note: $\hat{\beta}$ is the coefficient on the challenger's cost advantage. Outcomes are measured 30 days later. The sample contains $N = 57,989$ endpoint-pair-intermediary observations and 2,213 date clusters; standard errors are clustered by date and use the CR1 rank correction.

The estimates are large in the small set of endpoint markets and intermediary assets we can follow. They remain descriptive because the panel does not compare each market against a fixed set of feasible routes.

Architecture changes lack comparable route windows

	5%	10%	25%
Sustained entries	162 (0)	139 (0)	78 (0)
Substitution exits	79 (0)	69 (0)	54 (0)
Role disappearances	7 (3)	7 (3)	7 (3)
Immediate entry contrast	—	—	—
Immediate exit contrast	—	—	—

Every entry and exit coincides with a change in the comparison set or in whether the vehicle role exists. These windows cannot separate architecture from changes in active pairs and vehicles.

Note: Entries and substitution exits require the V4 share of an ordered endpoint-pair-intermediary series to cross the column threshold for three consecutive weeks on each side. Each entry and exit cell reports detected events, with usable non-overlapping 16-week windows in parentheses; role disappearances require the intermediary to be absent after the event.

Stablecoin dominance grew through changes in routed markets

- Stablecoins lead routed value in 2022 and again in 2025–2026, but fall behind native assets in 2023–2024.
- USDT drives the post-2024 rise in excess stablecoin intermediation; USDC changes little.
- Which endpoint pairs trade, how activity is distributed among them, and how often they route indirectly account for most of the aggregate gain. Direct vehicle switching within continuing markets contributes little.

Vehicle dominance is shaped by which markets form, not only by intermediary choice within established markets.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

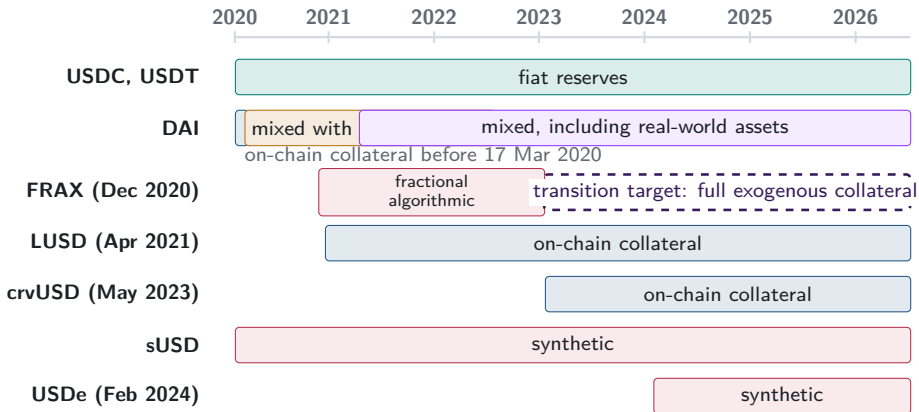
A1. One reconstructed route is one unit



one route unit
two swap legs

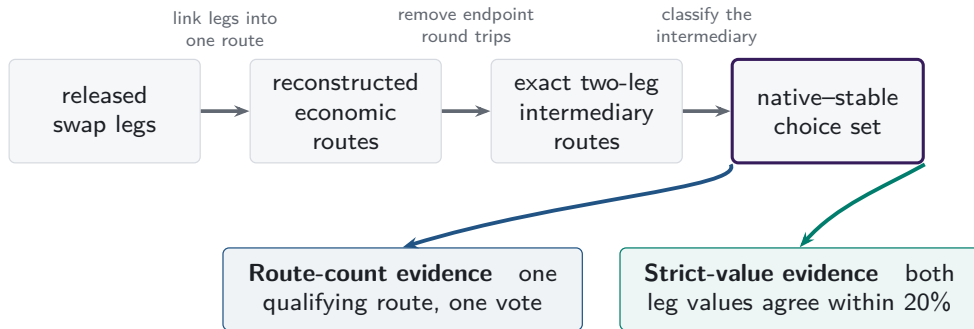
- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

A2. Stablecoin backing cannot be treated as fixed



Note: Bars begin at public launch or the sample start for older tokens. DAI's segments mark the backing regimes admitted by the classification; the dashed FRAX arrow is a governance target, not an observed collateral share.

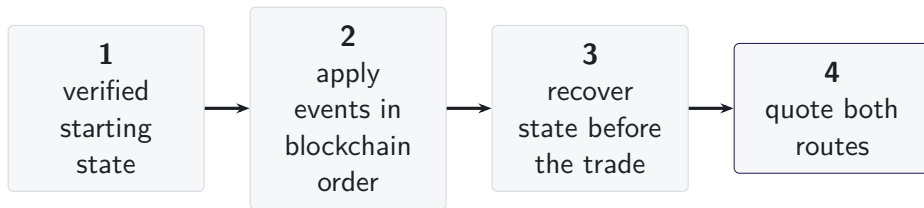
A3. One route universe supports two measurements



Topology-valid routes remain in count evidence when dollar support is unavailable.
Box widths do not represent attrition.

Missing dollar support narrows value-weighted evidence; it does not erase a topology-valid route from the count evidence.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

The common object is the state immediately before execution.
The fields required to recover that state vary with the pricing rule.

■ required state

□ hooks outside comparison

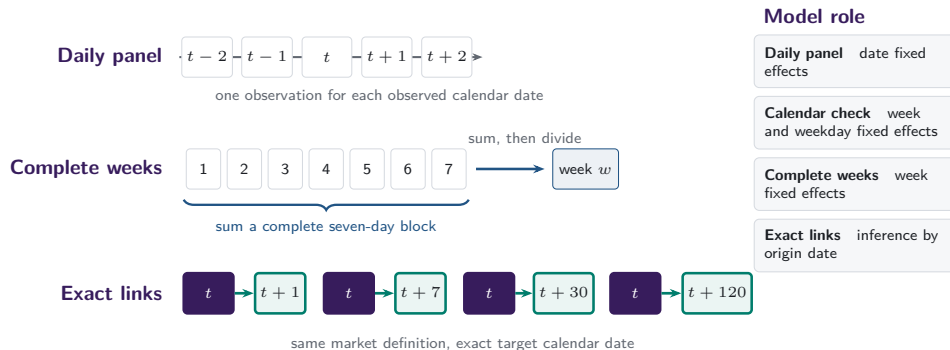
A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.



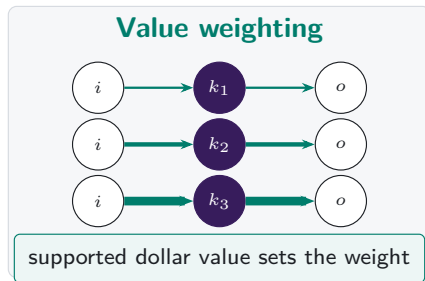
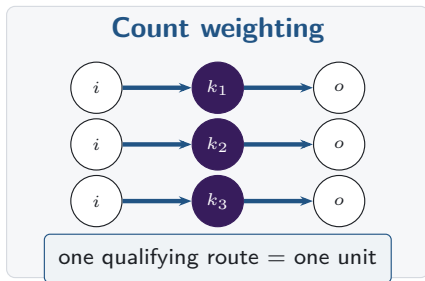
same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions



Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$I_{k,t}^N = \frac{n_{k,t}^I}{\sum_j n_{j,t}^I}$$

How often asset k sits inside a route.

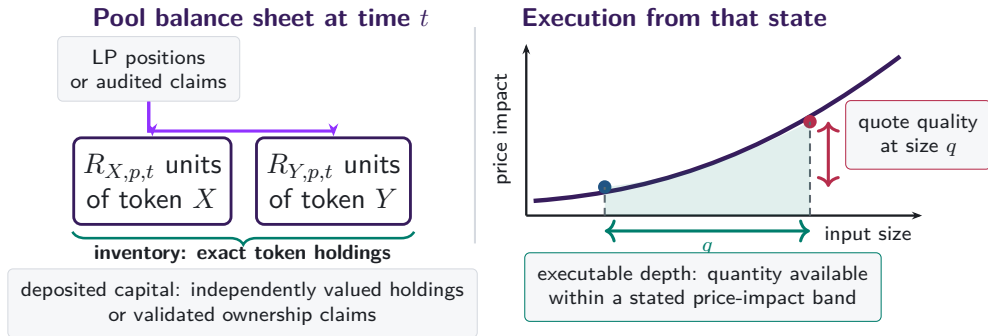
Endpoint share

$$E_{k,t}^N = \frac{n_{k,t}^E}{\sum_j n_{j,t}^E}$$

How often asset k is what traders buy or sell.

Excess use compares intermediation with endpoint demand on the same support.

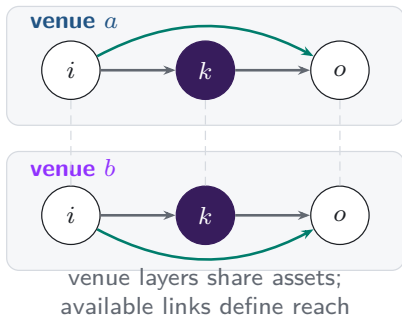
A10. Capital, inventory, and depth are different objects



Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Integration changes the routes that can be chosen

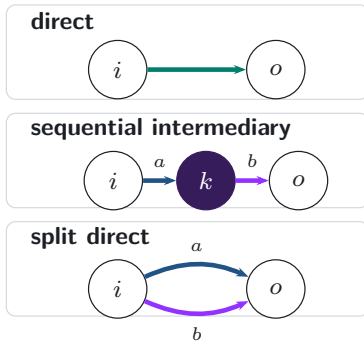
Feasible venue layers



realised
execution



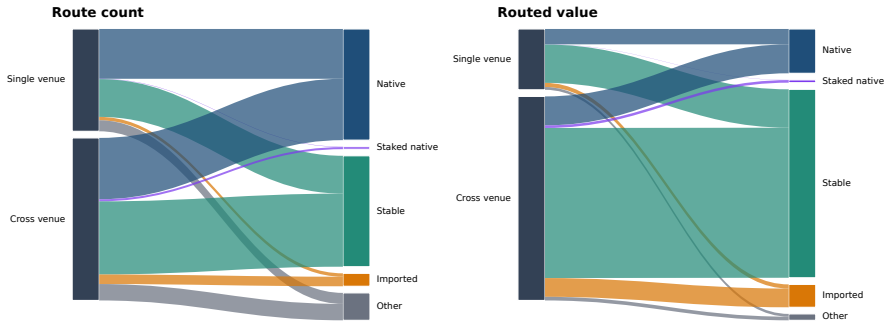
Realised route topology



The analysis therefore separates opportunity-set reach from the composition of realised routes.

A11b. Venue scope and vehicle type differ in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

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- Dowd and Greenaway (1993), *Economic Journal*
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A13. References: decentralised exchange design

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- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
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